Archer Case Study

Safe Autonomy with Runtime Assurance

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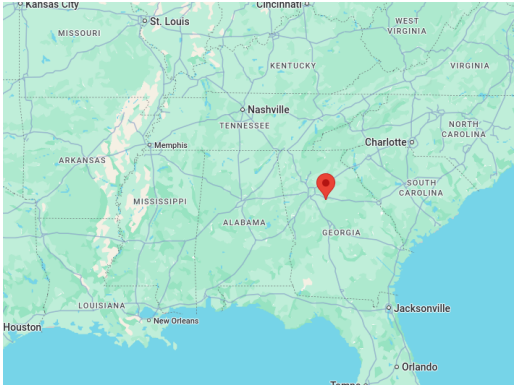
February 26, 2026

My Start: Covington, GA



Background

B.S. / M.S. / Ph.D. @ Georgia Tech — Aerospace & Robotics 2011–2021

Thesis: *Optimization-Based Approaches to Safety-Critical Control with Applications to Space Systems*

Intern @ ISAE-ENSMA — Quadcopter control 2015

GRA @ Georgia Tech — Safety verification on the Robotarium 2016–2021

Intern @ MIT Lincoln Lab — Neural network verification 2017–2018

Visiting Researcher @ Stanford — Collision-inclusive planning 2018

Intern @ NASA JPL — Spacecraft swarm control 2019

Visiting Student @ KAUST — Safe autonomy, control theory 2020

Intern @ AFRL — Safe spacecraft docking 2020

CEO & Co-Founder @ Pytheia — Camera perception for robots 2021–2024

CTO & Co-Founder @ Roostr — AI-native freight forwarder 2024–now

Template Slides

Reference examples for building new slides

Section Title

Brief subtitle or context

Content Slide Example

Key point or framing sentence at the top.

- First point with enough detail to stand alone
- Second point
  - Supporting detail
  - Supporting detail
- Third point

Equation Slide Example

The safety filter solves a quadratic program at each timestep:

Active Safety Filter

$$\min_{u} \quad \|u - u_{\text{des}}\|^2$$

$$\text{s.t.} \quad L_f h(x) + L_g h(x)\, u \geq -\alpha\big(h(x)\big)$$

- $h(x) \geq 0$ defines the safe set
- Minimally invasive: follows u_{des} when safe
- Small QP, solves in real time

Two-Column Slide Example

Left: Text or Math

- Explain the concept
- Key equations or points
- Connection to Archer

Right: Figure

Figure goes here

Full Image Slide Example

Full-width figure or screenshot

Brief caption or takeaway below the image.

Side-by-Side Image Example

Left image (4:3)

Right image (4:3)

Source citation here